PROJECT REPORT

OPTIMAL ENERGY MIX PLANNING FOR INDIA

Minimising the Total Cost of Energy Technologies to meet Total Demand in the background of Climate Commitments and Limited Capacities

MAY 1, 2026

DEVANSHU SHRIVASTAVA
MEng. ENERGY SYSTEMS

DR. DAVID W. COIT
PLANNING AND OPERATIONS ENGINEERING

RUTGERS
THE STATE UNIVERSITY
OF NEW JERSEY

TABLE OF CONTENTS

1. Introduction ……..……1

2. India's Current Energy Mix ……...…….……2

2.1. Installed Capacity Overview ……..2

2.2. Generation Mix and Trends ……....……3

2.3. Renewable Energy Targets and Policy Frameworks ……..……3

3. Problem Modelling …….…….4

3.1. What Is Energy Optimisation? ..……...…….…….4

3.2. Problem Description ……...…….……..4

3.2.1. Assumptions and Considerations ……..5

3.2.2. Variables' Definition ……..…6

3.2.3. Objective Function ……...……7

3.2.4. Decision Variables ……...……7

3.2.5. Constraints ……...…….8

3.2.5.1. Constraint 1: ……...……8

3.2.5.2. Constraint 2: ……...……9

3.2.5.3. Constraint 3 and 4: ..……..……10

3.2.5.4. Constraint 5: ……..……10

3.2.5.5. Constraint 6, 7 and 8: …..……...……10

3.2.5.6. Constraint 9: ……..……11

3.3 Optimisation Model ……...……12

3.4 Excel Spreadsheet ……...……13

3.4.1. Decision Variables in Excel ……13

3.4.2. Constraints in Excel ……13

3.4.2. Solver Parameters ……...……15

3.4.3. Solver Results ……...……...……15

4. Results ……...……16

4.1. Numerical Results ……16

4.2. General Description ……...……16

4.3. Constraint Sensitivity ……...……18

5. Conclusion ……..……20

1. INTRODUCTION

Energy is the lifeblood of economic development. For a country of India's scale — home to 1.4 billion people, the world's fastest-growing major economy, and a rapidly urbanising society — securing reliable, affordable, and clean energy is not merely a policy objective; it is an existential imperative. India is currently the third-largest producer and consumer of electricity globally, and its energy demand is projected to increase by 2 to 2.5 times by 2047 to meet the needs of a middle-income economy.

Historically, India's power sector was built on abundant domestic coal reserves. Coal-fired power plants provided the bulk of generation capacity and were instrumental in expanding electricity access across the country. However, this fossil-fuel-dependent model has come under increasing strain from multiple directions: rising carbon emissions and climate obligations, air quality degradation, reliance on imported petroleum and gas, and rapidly falling costs of renewable alternatives. At the same time, energy is at the heart of today’s geopolitical tensions, with traditional risks to fuel supply now joined by restrictions on critical mineral supplies. The electricity sector – so essential to modern economies – is also increasingly vulnerable to cyber, operational, and weather-related hazards. At the same time, the world remains thirsty for energy. Modern technologies are entering the system at a rapid pace, and renewables set new records for deployment in 2024 for the 23rd consecutive year. [1] Oil, natural gas, and coal consumption, and nuclear output all reached record highs. Driven by China and India, demand for coal has grown 50% faster than that of the next-fastest-growing fossil fuel, natural gas, a key reason energy-related emissions have continued to grow.

Energy mix optimisation planning is the systematic process of determining the combination of energy sources — coal, gas, solar, wind, hydro, nuclear, and others — that best satisfies defined objectives such as minimising total system cost, reducing carbon emissions, ensuring supply reliability, and improving energy security. In India's context, this is an extraordinarily complex problem, given the vast geographic diversity, the multiple stakeholders involved (central government, state governments, private developers, distribution companies), and the need to balance short-term economic considerations with long-term climate goals.

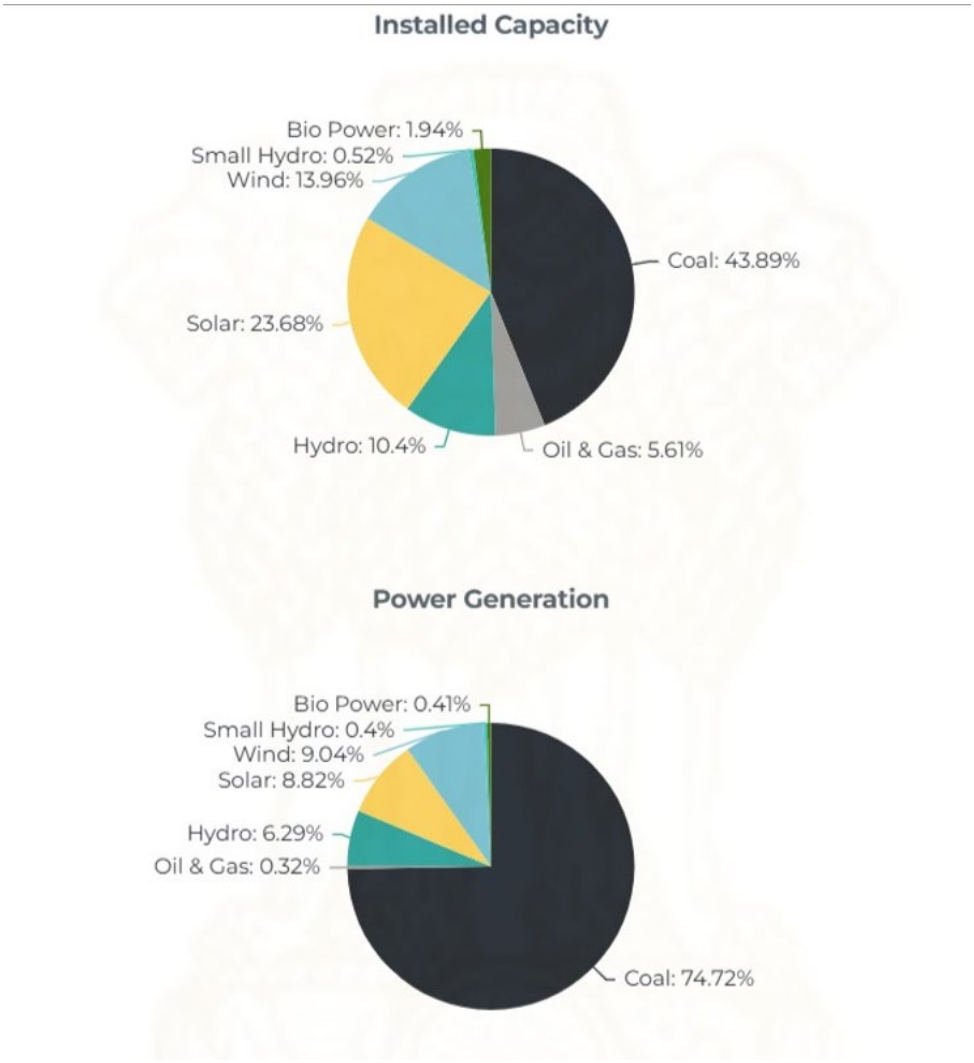
This project aims to determine the optimal combination of energy technologies — Solar, Wind, Hydro, Coal, and Gas — across ten major Indian states, with the objective of minimising the total capital cost of installed capacity while meeting each state's peak electricity demand. It models India's energy mix planning challenge as a Linear Programming (LP) resource-allocation problem and solves it using the Excel Solver Simplex LP method to guarantee a globally optimal solution. Ultimately, the project seeks to quantify not only the least-cost energy portfolio for India's transition, but also the economic premium associated with meeting its renewable energy targets — providing a data-driven foundation for evidence-based energy planning and policy decisions.

2. INDIA'S CURRENT ENERGY MIX

2.1. INSTALLED CAPACITY OVERVIEW

India's total installed power generation capacity crossed 505 GW as of late 2025, a landmark reflecting the rapid growth of the sector over the past decade. Of this, non-fossil fuel sources — including solar, wind, large hydro, small hydro, biomass, and nuclear — account for more than 259 GW, or over 50% of total installed capacity. This is a historic milestone, as it means India has already surpassed its 2030 NDC pledge of achieving 50% non-fossil capacity, five years ahead of the committed deadline.

The following figure summarises the installed capacity across major source categories as of 2025:



2.2. GENERATION MIX AND TRENDS

While the installed capacity picture is increasingly green, the generation mix — the actual electricity produced — remains heavily skewed toward fossil fuels. As of 2025, coal accounted for approximately 69% of total electricity generation, with total fossil fuel generation exceeding 72% of total output. This apparent paradox — more than 50% clean capacity but still over 70% fossil generation — reflects the lower capacity utilisation factor (CUF) of solar and wind relative to thermal plants that can run around the clock.

Nevertheless, the trend is clearly in favour of renewables. In FY 2024-25, solar generation grew by over 31 TWh and wind by over 25 TWh. Renewables accounted for 89% of all new capacity additions in FY 2024-25. Solar power alone added 23.83 GW in a single fiscal year, up from 15 GW the year before, signalling an accelerating deployment pace. Large hydro and solar have nearly converged in total generation, with solar rapidly catching up to hydro's historically dominant position in the renewable category.

India's electricity generation crossed 1,824 TWh in FY 2024-25 (utility only), growing at a compound annual rate of approximately 5% over the past decade. Demand peaked at a record 250 GW in May 2024, underscoring the urgency of continuing capacity expansion.

2.3. RENEWABLE ENERGY TARGETS AND POLICY FRAMEWORKS

India has set some of the world's most ambitious renewable energy targets, driven by both domestic development imperatives and international climate commitments. India ranks third globally in Renewable Energy Installed Capacity, according to the Renewable Energy Statistics 2026. India reached its highest-ever share of renewable energy in electricity generation, meeting 51.5% of the country’s total electricity demand. A total of 283.46 GW of capacity from non-fossil fuel sources has been installed in the country as of 2026. This includes 274.68 GW Renewable Energy (150.26 GW Solar Power, 56.09 GW Wind Power, 11.75 GW Bio Energy, 5.17 GW Small Hydro Power, 51.41 GW Large Hydro Power) and 8.78 GW Nuclear Power capacity.

India achieved the milestone of 50% of its cumulative electric power installed capacity from non-fossil fuel sources in June 2025, five years ahead of the 2030 target set under its Nationally Determined Contribution (NDC) to the Paris Agreement. For the period 2031-2035, the updated NDC include 500 GW of installed electricity capacity from non-fossil sources by 2030, reducing the emissions intensity of GDP by 47% from 2005 levels by 2035, increasing the share of non-fossil fuel-based installed electricity capacity to 60%, and creating a carbon sink of 3.5 to 4.0 billion tonnes of CO_2 equivalent through forest and tree cover by 2035.

3. PROBLEM MODELING

3.1. WHAT IS ENERGY MIX OPTIMISATION?

Energy mix optimisation (EMO), also known as Generation Expansion Planning (GEP), is the application of mathematical programming and simulation techniques to determine the optimal portfolio of electricity-generating technologies to deploy over a planning horizon. The 'optimality' is defined with respect to one or more objective functions, subject to a set of technical, economic, and policy constraints.

In the context of India's energy transition, EMO seeks to answer questions such as: What combination of solar, wind, hydro, storage, and thermal capacity minimises total system cost while reliably meeting demand? At what rate should coal capacity be retired to comply with emission targets without creating supply shortfalls? How much battery storage is needed to enable a high renewable penetration scenario? The answers have multi-trillion-rupee investment implications and shape the energy trajectory for decades.

3.2. PROBLEM DESCRIPTION

This project formulates India's energy mix planning problem as a linear programming (LP) resource-allocation problem. The central planning challenge is to determine the least-cost combination of 5 electricity-generating technologies — Solar, Wind, Hydro, Coal, and Gas — across its major states, while simultaneously satisfying peak demand requirements and policy-mandated constraints on emissions, renewables deployment, and grid reliability. The model is solved using Excel Solver (Simplex LP method), which guarantees a globally optimal solution for this class of problem. The study covers 10 representative high-demand states and 9 binding constraints, yielding 50 decision variables in total.

3.2.1. ASSUMPTIONS AND CONSIDERATIONS

To keep the model tractable within solver limits while preserving real-world fidelity, the following assumptions and simplifications have been made:

A. State Coverage: Ten large, energy-intensive states are selected as representatives of India's electricity sector — Uttar Pradesh (UP), Maharashtra (MH), Tamil Nadu (TN), Gujarat (GJ), Rajasthan (RJ), Madhya Pradesh (MP), Karnataka (KA), Andhra Pradesh (AP), Himachal Pradesh (HP), and Assam (AS). Together, these states account for the majority of India's total electricity demand.

B. Technology Coverage: Five technologies are modelled — Solar PV, Wind, Hydropower, Coal, and Natural Gas. These represent the dominant installed and planned capacity categories in India's current and near-future energy mix.

C. Linear Programming Scope: The objective function and all constraints are formulated as linear functions. This excludes non-linear effects such as economies of scale and unit-commitment start-up costs, which would require MILP or NLP formulations.

D. Capacity Factor as Efficiency Coefficient: Each technology's capacity factor (CF) — the fraction of installed capacity that actually generates electricity on average — is used as the efficiency coefficient in the demand satisfaction constraint. CFs are state- and technology-specific, sourced from NASA POWER and IEA data.

E. Static Single-Period Model: The model is a static snapshot optimisation (not a multi-period dynamic model). It determines the optimal installed capacity mix to meet peak demand in a single representative planning period.

F. Cost Uniformity Across States: The capital cost (cost of installation) per MW is assumed to be uniform across states for the same technology. This is a simplification; in practice, costs vary by terrain, grid proximity, and labour.

G. Coal Transition Sensitivity: Both a ceiling cap (maximum coal allowed per state, derived from climate targets) and a floor (minimum coal required, reflecting just transition concerns for coal-dependent communities and existing infrastructure) are imposed on coal capacity.

H. Demand Figure: Peak demand figures are sourced from CEA FY 2023-24 data and rounded to the nearest 500 MW for modelling simplicity.

3.2.2. VARIABLES' DEFINITION

The following types of variables have been used in the optimisation modelling. This notation is used consistently throughout the model formulation.

Definitions	***Variables***
Amount of installed capacity (MW) in state i using energy source *j* [Decision Variable]	$x_{i,j}$
State index: {UP, MH, TN, GJ, RJ, MP, KA, AP, HP, AS}	i
Technology index: {Solar, Wind, Hydro, Coal, Gas}	j
Capital cost per MW of technology *j* (USD/MW)	c_j
Peak electricity demand of state *i* (MW)	D_i
Capacity factor of technology *j* in state *i* (dimensionless, 0 to 1)	$\alpha_{i,j}$
Maximum coal capacity allowed in state *i* (MW) [Coal ceiling]	C_i^{max}
Minimum coal capacity required in state *i* (MW) [Coal floor]	C_i^{min}
Minimum dispatchable capacity required in state $i = 0.25 \times D_i$ (MW)	S_i
Hydro dispatchability coefficient applied in the grid stability constraint	β
Minimum solar capacity required in state *i* (MW)	$R_{i,solar}^{min}$
Minimum wind capacity required in state *i* (MW)	$R_{i,wind}^{min}$
Minimum hydro capacity required in state *i* (MW)	$R_{i,hydro}^{min}$
Maximum installable capacity for state *i* and technology *t* (MW) [Land resource cap]	$Cap_{i,t}$

3.2.3. OBJECTIVE FUNCTION

The objective is to minimise the total capital cost of the energy mix across all states and technologies. The cost per unit of each technology is uniform across, so the objective collapses to a sum over all 50 decision variables weighted by their respective technology cost:

Technology (*j*)	Cost per MW Installed (USD/MW)
Solar	55,000
Wind	65,000
Hydro	80,000
Coal	45,000
Gas	30,000

Gas has the lowest capital cost per MW but is constrained by grid reliability and environmental requirements. Coal has lower capital costs than renewables but is constrained by climate commitments. Hydro has the highest unit cost but provides critical baseload and dispatchability.

The objective function is:

$$min\ Z = \Sigma_i \Sigma_j \left(c_i . x_{i,j} \right)$$

3.2.4. DECISION VARIABLES

There are 50 decision variables in total — one for each state-technology combination (10 states multiplied by 5 technologies). Each variable, $x_{i,j}$, represents the amount of installed generating capacity (in MW) recommended for state i using technology j. All decision variables are constrained to be non-negative ($x_{i,j} \geq 0$).

ENERGY PLANNING FOR INDIA						
Decision Variables						
		Solar	Wind	Hydro	Coal	Gas
	Uttar Pradesh	0	0	0	0	0
	Maharashtra	0	0	0	0	0
	Tamil Nadu	0	0	0	0	0
	Gujarat	0	0	0	0	0
	Rajasthan	0	0	0	0	0
	Madhya Pradesh	0	0	0	0	0
	Karnataka	0	0	0	0	0
	Andhra Pradesh	0	0	0	0	0
	Himachal Pradesh	0	0	0	0	0
	Assam	0	0	0	0	0

3.2.5. CONSTRAINTS

The model incorporates 9 constraints across 5 categories: demand requirements, physical feasibility, climate policy, grid reliability, and land resource limits. Together they constitute 10 + 10 + 10 + 10 + 10 + 10 + 10 + 10 + 20 = 100 individual constraint inequalities across the 10 states. Each of the constraints is described as follows:

3.2.5.1. CONSTRAINT 1: DEMAND SATISFACTION

Each state must generate at least enough electricity to meet its peak demand. The effective output of each technology is scaled by its capacity factor $\alpha_{ij.}$

$$\Sigma_j \alpha_{i,j}.x_{i,j} \geq D_i \forall\, i$$

#	State	Peak Demand D_i (MW)
1	Uttar Pradesh	28,000
2	Maharashtra	32,000
3	Tamil Nadu	20,000
4	Gujarat	22,000
5	Rajasthan	19,000

6	Madhya Pradesh	15,000
7	Karnataka	14,500
8	Andhra Pradesh	13,000
9	Himachal Pradesh	2,000
10	Assam	2,500

3.2.5.2. CONSTRAINT 2: CAPACITY FACTORS

The capacity factor $\alpha_{i,j}$ is the fraction of installed capacity that actually generates electricity on average, accounting for solar irradiance, wind speed, hydro resource, and thermal plant availability. These are state- and technology-specific values derived from NASA POWER data and IEA reports:

State	Solar	Wind	Hydro	Coal	Gas
Uttar Pradesh	0.20	0.10	0.10	0.70	0.85
Maharashtra	0.22	0.25	0.15	0.70	0.85
Tamil Nadu	0.20	0.30	0.10	0.70	0.85
Gujarat	0.23	0.28	0.08	0.70	0.85
Rajasthan	0.25	0.15	0.05	0.70	0.85
Madhya Pradesh	0.22	0.20	0.12	0.70	0.85
Karnataka	0.22	0.25	0.30	0.70	0.85
Andhra Pradesh	0.22	0.28	0.15	0.70	0.85
Himachal Pradesh	0.18	0.12	0.45	0.70	0.85
Assam	0.17	0.10	0.35	0.70	0.85

3.2.5.3. CONSTRAINTS 3 AND 4: COAL GENERATION BOUNDS

Coal is doubly bound. The ceiling cap (Constraint 3) reflects India's NDC commitment to limit coal's share in the energy mix, ensuring the LP solution is consistent with the trajectory toward 60% non-fossil capacity by 2035. The floor (Constraint 4) reflects the just-transition imperative: coal-dependent communities and existing infrastructure require a minimum level of coal generation to be maintained during the transition period.

$$C_i^{min} \leq x_{i,coal} \leq C_i^{max} \forall i$$

The ceiling C_i^{max} is set at approximately 75% of each state's current coal capacity (reflecting the NDC-aligned phase-down trajectory). The floor C_i^{min} represents a minimum just-transition threshold per state.

3.2.5.4. CONSTRAINT 5: GRID STABILITY (DISPATCHABLE POWER)

Renewable energy sources (Solar and Wind) cannot guarantee a round-the-clock supply. To ensure grid reliability — including nights and calm days — each state must maintain a minimum level of dispatchable (always-on) capacity equal to 25% of its peak demand. The dispatchable sources in this model are Coal (fully dispatchable), Gas (fully dispatchable), and Hydro (partially dispatchable, weighted by its capacity factor β):

$$x_{i,coal} + x_{i,gas} + \beta \cdot x_{i,hydro} \geq S_i \forall i$$

where $S_i = 0.25 \times D_i$

3.2.5.5. CONSTRAINTS 6, 7 AND 8: MINIMUM RENEWABLE REQUIREMENTS

To align with India's renewable energy policy targets (500 GW non-fossil by 2030, NDC renewables mandate), minimum floor values for Solar, Wind, and Hydro capacity are imposed on each state. These floors are derived from state-level renewable energy targets published by MNRE:

$$x_{i,solar} \geq R_{i,solar}^{min} \quad (Constraint6)$$

$$x_{i,wind} \geq R_{i,wind}^{min} \quad (Constraint7)$$

$$x_{i,hydro} \geq R_{i,hydro}^{min} \quad (Constraint8)$$

State	Min Solar (MW)	Min Wind (MW)	Min Hydro (MW)
Uttar Pradesh	22,000	2,000	500
Maharashtra	15,000	5,000	1,000
Tamil Nadu	10,000	12,000	500
Gujarat	30,000	15,000	200
Rajasthan	40,000	8,000	200
Madhya Pradesh	10,000	3,000	500
Karnataka	12,000	6,000	2,000
Andhra Pradesh	12,000	8,000	500
Himachal Pradesh	800	400	4,000
Assam	500	200	1,500

3.2.5.6. CONSTRAINT 9: LAND RESOURCE LIMITS

Solar and Wind installations require significant land. Maximum installable capacity for Solar and Wind in each state is capped based on MNRE resource potential assessments and available land, ensuring the LP solution is physically feasible:

$$x_{r,t} \leq Cap_{r,t} \; \forall r, \forall t \in Solar, Wind$$

State	Max Solar (MW)	Max Wind (MW)
Uttar Pradesh	40,000	5,000
Maharashtra	30,000	10,000

Tamil Nadu	20,000	35,000
Gujarat	40,000	30,000
Rajasthan	75,000	10,000
Madhya Pradesh	35,000	8,000
Karnataka	25,000	15,000
Andhra Pradesh	30,000	20,000
Himachal Pradesh	2,000	1,000
Assam	3,000	1,000

3.3. OPTIMIZATION MODEL

The complete LP model is stated below in standard form. The model has 50 decision variables (all non-negative), 1 objective function, and 9 constraint categories (approx. 100 individual constraint inequalities across 10 states).

SETS:

$$i \in \{UP, MH, TN, GJ, RJ, MP, KA, AP, HP, AS\}$$
$$j \in \{Solar, Wind, Hydro, Coal, Gas\}$$

DECISION VARIABLES:

$$x_{i,j} \geq 0 \; \forall i, j$$

OBJECTIVE FUNCTION:

$$minZ = \Sigma_i \Sigma_j \left(c_j \cdot x_{i,j}\right)$$

$$= 55{,}000 \cdot \Sigma_i x_{i,Sol} + 65{,}000 \cdot \Sigma_i x_{i,Win} + 80{,}000 \cdot \Sigma_i x_{i,Hyd} + 45{,}000 \cdot \Sigma_i x_{i,Coal} + 30{,}000 \cdot \Sigma_i x_{i,Gas}$$

SUBJECT TO:

$$\alpha_{i,Sol} \cdot x_{i,Sol} + \alpha_{i,Win} \cdot x_{i,Win} + \alpha_{i,Hyd} \cdot x_{i,Hyd} + 0.70 \cdot x_{i,Coal} + 0.85 \cdot x_{i,Gas} \geq D_i$$

$$x_{i,coal} \leq C_i^{max}$$

$$x_{i,coal} \geq C_i^{min}$$

$$x_{i,coal} + x_{i,gas} + \beta \cdot x_{i,hydro} \geq 0.25 \cdot D_i$$

$$x_{i,solar} \geq R_{i,solar}^{min}$$

$$x_{i,wind} \geq R_{i,wind}^{min}$$

$$x_{i,hydro} \geq R_{i,hydro}^{min}$$

$$x_{i,solar} \leq Cap_{i,solar}$$

$$x_{i,wind} \leq Cap_{i,wind}$$

$$x_{i,j} \geq 0$$

3.4. EXCEL SPREADSHEET

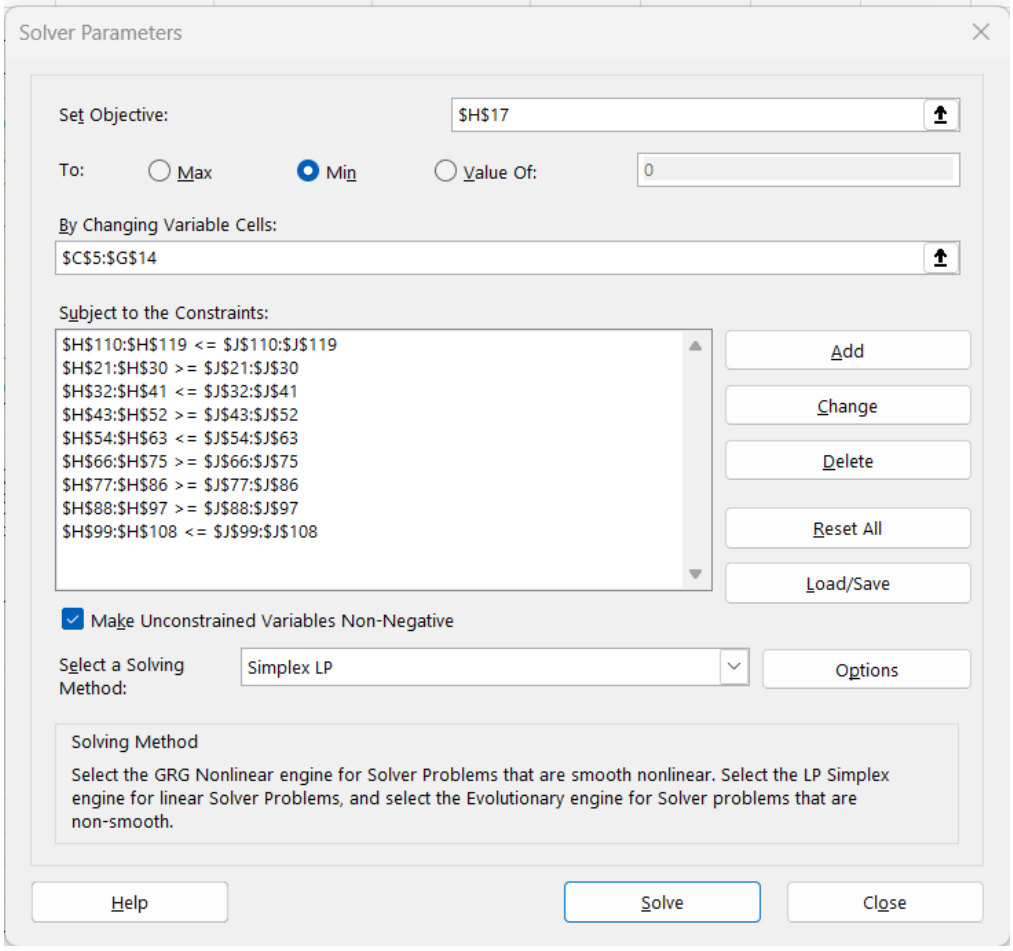
3.4.1. DECISION VARIABLES IN EXCEL

	A	B	C	D	E	F	G
1	**ENERGY PLANNING FOR INDIA**						
2							
3	**Decision Variables**						
4			Solar	Wind	Hydro	Coal	Gas
5		Uttar Pradesh	0	0	0	0	0
6		Maharashtra	0	0	0	0	0
7		Tamil Nadu	0	0	0	0	0
8		Gujarat	0	0	0	0	0
9		Rajasthan	0	0	0	0	0
10		Madhya Pradesh	0	0	0	0	0
11		Karnataka	0	0	0	0	0
12		Andhra Pradesh	0	0	0	0	0
13		Himachal Pradesh	0	0	0	0	0
14		Assam	0	0	0	0	0
15							

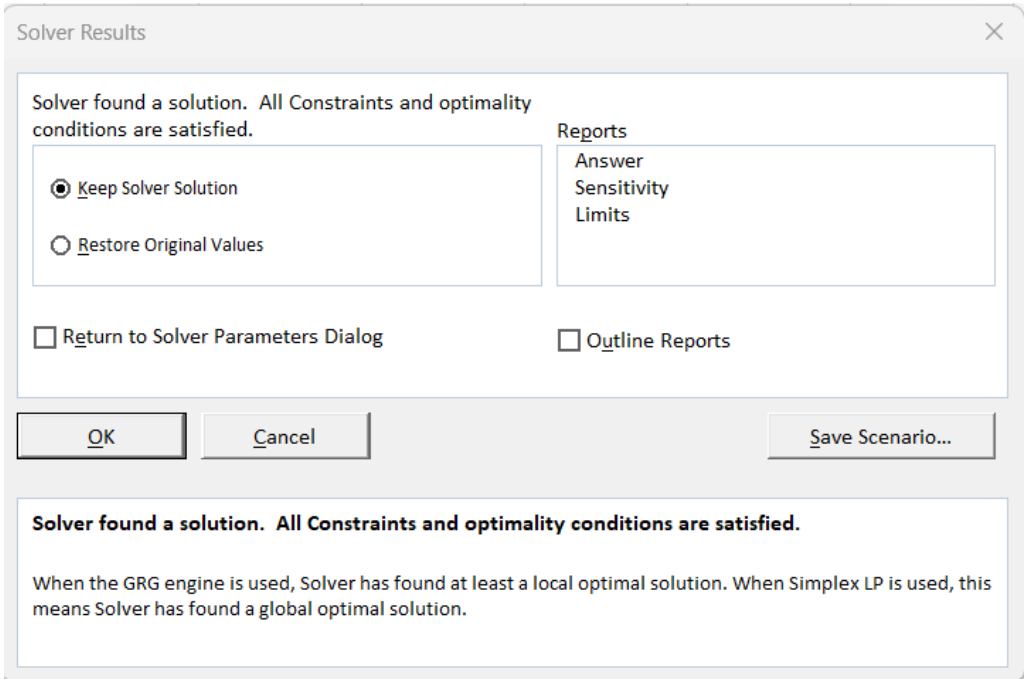
3.4.2. CONSTRAINTS IN EXCEL

4			Solar	Wind	Hydro	Coal	Gas			
64										
65	Renewables floor minimum									
66	For solar	Uttar Pradesh	0					0	>=	22,000
67		Maharashtra	0					0	>=	15,000
68		Tamil Nadu	0					0	>=	10,000
69		Gujarat	0					0	>=	30,000
70		Rajasthan	0					0	>=	40,000
71		Madhya Pradesh	0					0	>=	10,000
72		Karnataka	0					0	>=	12,000
73		Andhra Pradesh	0					0	>=	12,000
74		Himachal Pradesh	0					0	>=	800
75		Assam	0					0	>=	500
76										
77	For wind	Uttar Pradesh		0				0	>=	2,000
78		Maharashtra		0				0	>=	5,000
79		Tamil Nadu		0				0	>=	12,000
80		Gujarat		0				0	>=	15,000
81		Rajasthan		0				0	>=	8,000
82		Madhya Pradesh		0				0	>=	3,000
83		Karnataka		0				0	>=	6,000
84		Andhra Pradesh		0				0	>=	8,000
85		Himachal Pradesh		0				0	>=	400
86		Assam		0				0	>=	200
87										
88	For Hydro	Uttar Pradesh			0			0	>=	500
89		Maharashtra			0			0	>=	1,000
90		Tamil Nadu			0			0	>=	500
91		Gujarat			0			0	>=	200
92		Rajasthan			0			0	>=	200
93		Madhya Pradesh			0			0	>=	500
94		Karnataka			0			0	>=	2,000
95		Andhra Pradesh			0			0	>=	500
96		Himachal Pradesh			0			0	>=	4,000
97		Assam			0			0	>=	1,500
98										

3.4.3. SOLVER PARAMETERS



3.4.4. SOLVER RESULTS



4. RESULTS

4.1. NUMERICAL RESULTS

The Excel Solver (Simplex LP) confirmed a globally optimal solution satisfying all constraints and optimality conditions. The minimum total capital cost of the energy mix across 10 states is:

Optimal Objective Function Value: $Z^* = 17{,}676{,}791{,}176 (\approx 17.68\ Billion)$

The Technology and state-wise cost breakdown of the optimal solution is shown below:

ENERGY PLANNING FOR INDIA

Decision Variables							
		Solar	Wind	Hydro	Coal	Gas	
	Uttar Pradesh	22000	2000	500	5400	23023.52941	
	Maharashtra	15000	5000	1000	4800	28164.70588	
	Tamil Nadu	10000	12000	500	2400	14905.88235	
	Gujarat	30000	15000	200	2700	10581.17647	
	Rajasthan	40000	8000	200	2100	7435.294118	
	Madhya Pradesh	10000	3000	500	2550	12182.35294	
	Karnataka	12000	6000	2000	1200	10494.11765	
	Andhra Pradesh	12000	8000	500	2250	7611.764706	
	Himachal Pradesh	800	400	4000	0	9.411764706	
	Assam	500	200	1500	150	2076.470588	
Objective Function							
Minimize total cost		8376500000	3874000000	872000000	1059750000	3494541176	17676791176
Cost of installation		55,000	65,000	80,000	45,000	30,000	

4.2. GENERAL DESCRIPTION

The optimal solution produced by the LP model reveals a geographically differentiated energy mix that closely reflects the physical resource endowments, climate constraints, and demand characteristics of each state. Several distinctive patterns emerge:

A. Solar Dominance in Desert and Semi-Arid States: Rajasthan emerges as the solar champion, with 40,000 MW of solar capacity — the maximum land-use cap for the state. This is entirely consistent with Rajasthan's geography: the Thar Desert offers the highest solar irradiance in India (a model capacity factor of 0.25, the highest among all states), vast uninhabited land, and minimal competing land uses. Gujarat follows with 30,000 MW of solar, also near its resource cap, reflecting its strong solar regime and flat terrain.

B. Hydro Dominance in Himalayan States: Himachal Pradesh presents the starkest contrast. The model allocates 4,000 MW of hydro — the dominant share of its energy

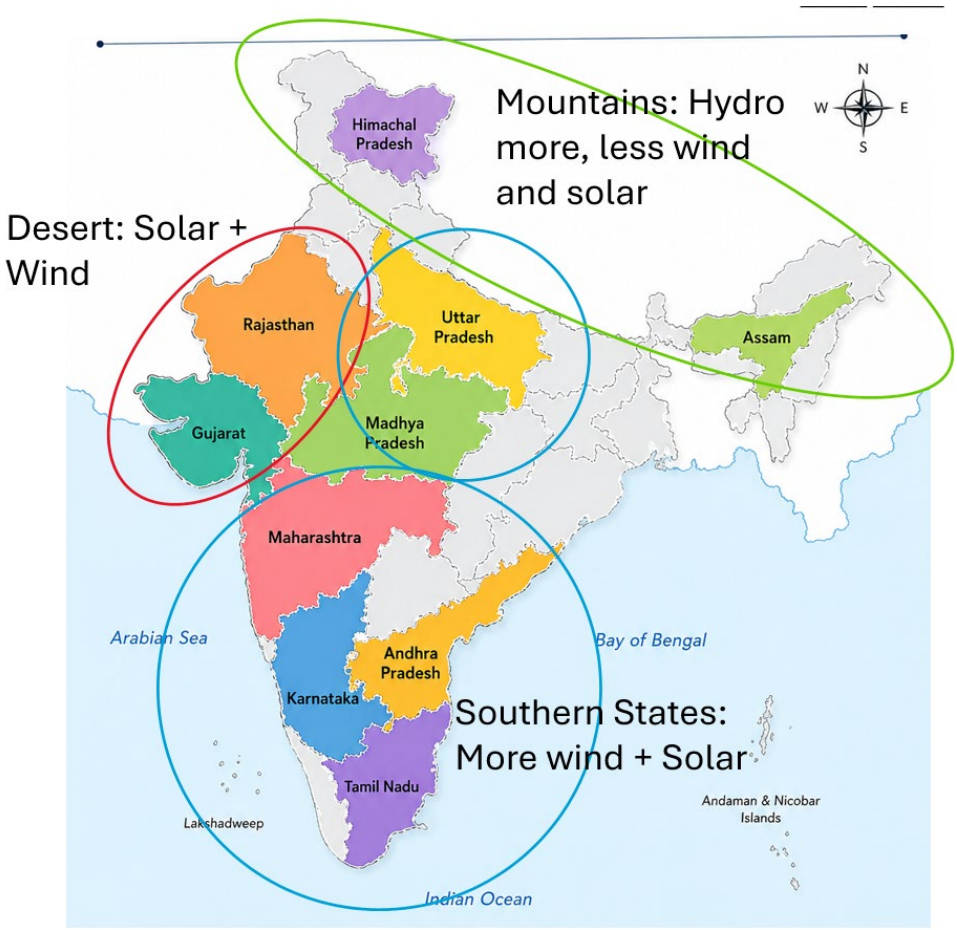
mix — while coal allocation is essentially zero (approximately 11 MW, well below even the minimum just-transition floor of 150 MW, indicating the floor is non-binding here because hydro already more than satisfies the dispatchable requirement). This reflects Himachal Pradesh's Himalayan river systems, which provide the highest hydro capacity factor (0.45) of any state in the model, and its complete absence of coal resources.

C. Wind Leadership in Coastal and Southern States: Tamil Nadu receives the highest wind allocation of any state at 12,000 MW — reflecting India's best onshore wind corridor along its coastline, modelled with a capacity factor of 0.30. Maharashtra and Karnataka also receive significant wind allocations (5,000 and 6,000 MW, respectively), consistent with their coastal exposure and the Western Ghats' orographic wind effects.

D. Gas as the Gap-Filler in High-Demand, Low-Renewable States: The most striking result is the very high gas allocation in Uttar Pradesh: 23,023 MW — the highest gas capacity of any state. The model's logic is transparent: UP has the highest peak demand (28,000 MW) among all states, but relatively low renewable capacity factors (solar CF = 0.20, wind CF = 0.10) and limited hydro potential. Coal is already at its floor minimum (5,400 MW—the minimum just transition threshold). With the minimum renewable requirements binding, gas — which has the lowest capital cost ($30,000/MW) — becomes the cheapest feasible gap-filler to meet the remaining demand. Maharashtra similarly receives 28,164 MW of gas because it has both high demand and moderate renewable CFs, making coal near its ceiling and gas the least-cost complement.

E. Andhra Pradesh and Karnataka — Zero Gas: Andhra Pradesh and Karnataka receive no gas in the optimal solution. This reflects the model's preference for coal over gas, where coal's capital cost ($45,000/MW) competes favourably with gas ($30,000/MW) after accounting for the higher gas capacity factor (0.85 vs coal's 0.70). For states where coal is not at its ceiling, and renewable CFs are adequate, coal provides a cheaper baseload complement than gas.

	Solar	Wind	Hydro	Coal	Gas
Uttar Pradesh	22000	2000	500	5400	23023.52941
Maharashtra	15000	5000	1000	4800	28164.70588
Tamil Nadu	10000	12000	500	2400	14905.88235
Gujarat	30000	15000	200	2700	10581.17647
Rajasthan	40000	8000	200	2100	7435.294118
Madhya Pradesh	10000	3000	500	2550	12182.35294
Karnataka	12000	6000	2000	1200	10494.11765
Andhra Pradesh	12000	8000	500	2250	7611.764706
Himachal Pradesh	800	400	4000	0	9.411764706
Assam	500	200	1500	150	2076.470588



4.3. CONSTRAINT SENSITIVITY:

The Cost of Climate Commitments: A revealing sensitivity analysis was performed by removing each constraint and observing the change in the optimal cost. The results quantify precisely which constraints are 'binding' (actively shaping the optimal solution) and by how much.

The most important finding from the sensitivity analysis is that the minimum renewables constraint carries a shadow cost of $11.28 billion — meaning that requiring states to meet minimum solar, wind, and hydro floors adds $11.28 billion to the system cost compared to an unconstrained pure cost-minimisation solution. Without the renewables mandate, the LP would default almost entirely to the cheapest option: gas and coal. This $11.28 billion represents the quantified economic cost of India's climate commitments — the premium the country pays to achieve its NDC and 500 GW clean energy targets. This finding is a powerful validation of the model and a key policy insight: the energy transition is feasible but not free, and the renewable minimum constraints are the most binding (and most expensive) constraints in the system.

The coal ceiling cap and the land resource cap are both non-binding at the optimum, meaning that even without these constraints, the model would not choose to build more coal or renewable capacity than currently permitted. This is because: (a) gas is already cheaper than coal at the margin in most states; and (b) the renewable minimum constraints are already forcing more solar and wind than the LP would otherwise choose. The grid stability constraint is similarly non-binding — the combination of coal minimums and gas allocations already exceeds the 25% dispatchable requirement in all states.

Constraint Removed	New Minimum Cost	Cost Difference
Demand not required to be met	$14.68 Billion	−$3.00 billion
Without Coal Ceiling Cap	$17.68 Billion	$0 (Non-binding)
Without Coal Floor (Just Transition)	$17.20 Billion	−$480 million
Without Min Conventional Sources (Grid)	$17.68 Billion	$0 (Non-binding)
Without Minimum Renewables Constraint	$6.40 Billion	−$11.28 billion
Without the Land Resource Cap	$17.68 Billion	$0 (Non-binding)

5. CONCLUSION

This project demonstrates the power of Linear Programming as a practical, computationally tractable tool for national-level energy mix planning. By formulating India's generation planning problem as a 50-variable LP with 9 constraint categories, and solving it using Excel Solver's Simplex LP method, the study arrives at a globally optimal capacity allocation that is simultaneously cost-minimising, demand-satisfying, and policy-compliant.

The central finding is that the optimal energy mix naturally mirrors India's geographic and resource reality. The model does not need to be told that Rajasthan should be a solar powerhouse or that Himachal Pradesh should rely on hydro — these emerge organically from the capacity factors, land constraints, and cost coefficients embedded in the LP. This is the fundamental strength of optimisation-based planning: it encodes physical reality and policy requirements as constraints and lets the mathematics find the best feasible solution.

Several specific insights stand out. Gas emerges as the dominant residual baseload source in high-demand, low-renewable-CF states like Uttar Pradesh and Maharashtra — not because gas is environmentally desirable, but because it is the cheapest technology that can satisfy both the dispatchable power requirement and the remaining demand gap after renewables' minimums are met. This is a direct consequence of the LP's cost-minimisation logic and highlights a real planning challenge: as gas prices are volatile and India imports most of its natural gas, the financial risk embedded in this optimal solution warrants further analysis beyond static capital cost minimisation.

The constraint sensitivity analysis delivers perhaps the most policy-relevant insight of the entire study: India's climate commitments (minimum renewables requirements) add $11.28 billion to the system cost — a premium of approximately 64% above the pure least-cost solution of $6.40 billion. This is not a criticism of the policy; it is a measurement of it. Every dollar of these premium purchases results in carbon emission reductions, improvements in energy security through domestic renewable resources, and progress toward the 500 GW non-fossil target. Understanding this trade-off quantitatively is essential for policymakers, investors, and regulators.

The model has several limitations that point to opportunities for future research. First, it is a static single-period model. It does not capture the temporal dynamics of renewable intermittency (solar at night, wind on calm days) or the multi-period investment planning problem of phased capacity additions. Second, the cost structure is simplified — it uses only capital costs and ignores fuel costs, O&M, transmission, and social costs. Third, the LP formulation cannot capture integer decisions (e.g., the discrete unit size of power plants) or non-linear cost curves. Future work could extend this to a multi-period MILP with stochastic renewable generation, incorporate Levelized Cost of Electricity (LCOE) instead of pure capital cost, and add storage (battery and pumped

hydro) as additional decision variables. Such extensions would bring the model closer to the CEA's Optimal Generation Mix 2030 framework and further strengthen its value as a planning tool.

In conclusion, the LP model successfully demonstrates that a cost-optimal, climate-compliant energy mix for India's 10 major states can be systematically determined and that the solution is geographically coherent, policy-consistent, and computationally validated. The optimal total cost of $17.68 billion for 362,835 MW of new capacity across ten states provides a rigorous, quantitative baseline for India's energy transition planning.